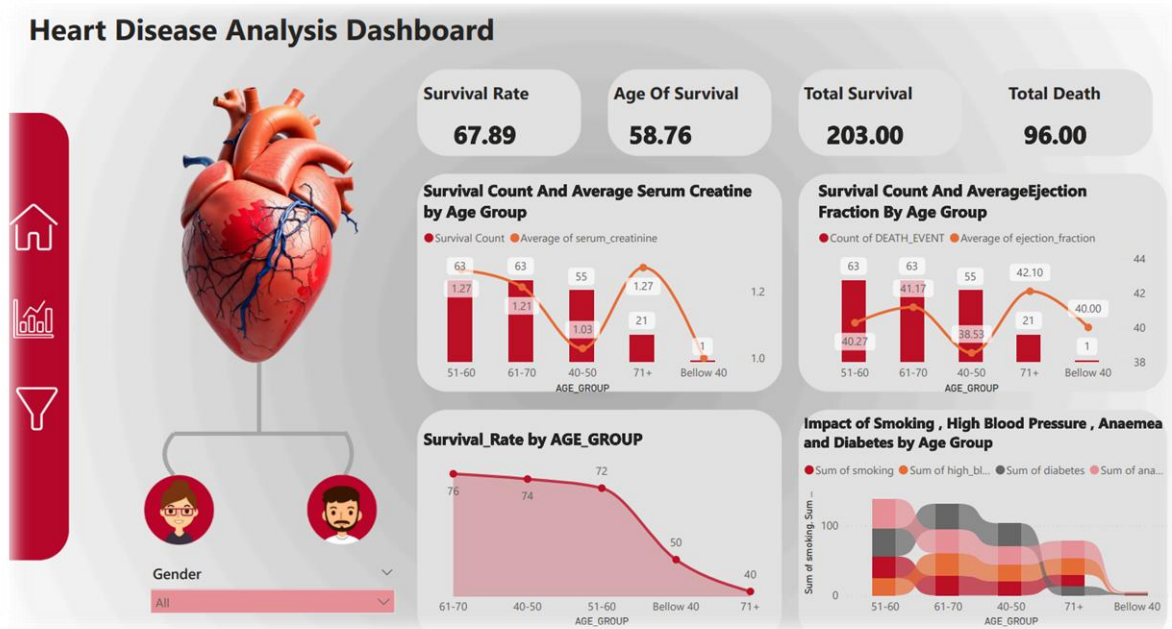


DAY 18 POWER BI PRACTICE

Heart Disease Analysis Dashboard Report



Project Overview

The **Heart Disease Analysis Dashboard** is an interactive Power BI visualization designed to explore and analyze patient health data related to **heart disease survival and mortality**. The dashboard provides actionable insights about survival rates, age group comparisons, and the impact of health conditions (like diabetes, smoking, high blood pressure, and anaemia) on survival outcomes.

Objectives

1. To analyze **heart disease survival rates** among different **age groups**.
2. To determine the **relationship between serum creatinine and survival**.
3. To study how **ejection fraction** and other factors affect death events.
4. To understand the **impact of risk factors** (like smoking, diabetes, anaemia, and high blood pressure).
5. To create an **interactive and visually informative dashboard** using Power BI.

Dataset Description

The dataset used for this analysis contains medical records of patients suffering from heart failure. Each record includes demographic details and medical measurements.

Feature Name	Description
Age	Age of the patient
Gender	Male/Female
Serum Creatinine	Level of serum creatinine in blood (mg/dL)
Ejection Fraction	Percentage of blood leaving the heart each time it contracts
Smoking	1 = Smoker, 0 = Non-smoker
Diabetes	1 = Diabetic, 0 = Non-diabetic
Anaemia	1 = Anaemic, 0 = Normal
High Blood Pressure	1 = Yes, 0 = No
Death Event	1 = Died, 0 = Survived
Age Group	Categorized as Below 40, 40–50, 51–60, 61–70, 71+

 Dashboard Components and Visuals:

1. Survival Statistics Cards

- **Survival Rate:** 67.89%
- **Average Age of Survival:** 58.76 years
- **Total Survival:** 203
- **Total Deaths:** 96

These KPI cards summarize the core metrics, offering an instant overview of heart disease outcomes.

2. Survival Count and Average Serum Creatinine by Age Group

- **Chart Type:** Clustered Column and Line Chart
- **Insights:**
 - Serum creatinine levels increase with age.
 - Highest survival counts are observed in the 51–70 age range.
 - Younger patients (below 40) have fewer cases but relatively better health indicators.

3. Survival Count and Average Ejection Fraction by Age Group

- **Chart Type:** Column and Line Combination
- **Insights:**
 - Ejection fraction decreases as age increases, indicating weaker heart function in older groups.
 - Death events are highest among 51–70-year-old patients.
 - The lowest average ejection fraction (~38.5%) is seen in the 40–50 group.

4. Survival Rate by Age Group

- **Chart Type:** Area Chart
- **Insights:**
 - The survival rate drops sharply after age 60.

- Highest survival is seen among the 40–50 and 51–60 groups.
- Only 40% of patients aged 71+ survived.

5. 📊 Impact of Smoking, High Blood Pressure, Anaemia, and Diabetes

- **Chart Type:** Stacked Bar (or Ribbon) Chart
- **Insights:**
 - Smoking and high blood pressure are most common in the 51–70 age groups.
 - Diabetes and anaemia show a clear increase with age.
 - The combined effect of multiple health risks increases mortality in older age groups.

Measures and DAX Formulas Used

Measure Name	DAX Formula (Example)
Survival Rate	<code>DIVIDE(CALCULATE(COUNTROWS(Data), Data[Death_Event] = 0), COUNTROWS(Data)) * 100</code>
Average Age of Survival	<code>AVERAGEX(FILTER(Data, Data[Death_Event] = 0), Data[Age])</code>
Total Survival	<code>CALCULATE(COUNTROWS(Data), Data[Death_Event] = 0)</code>
Total Death	<code>CALCULATE(COUNTROWS(Data), Data[Death_Event] = 1)</code>

Insights & Conclusion

- Survival rates **decrease sharply with increasing age**.
- **Serum creatinine** is a strong indicator of poor survival chances.
- Patients with **multiple conditions** (smoking + diabetes + high BP) have a significantly higher mortality rate.
- Preventive care in the **40–60 age group** could reduce death events drastically.